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Open access, generative artificial intelligence, and the criminology evidence base

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ABSTRACT

Open access (OA) and generative artificial intelligence (genAI) shape how researchers, practitioners, and policymakers discover and synthesize criminological knowledge. We examined how deep-research tools from three popular genAI platforms handle criminology literature reviews. The systems vary considerably in quality: some produce plausible reports with unreliable citations, while others generate structured reviews drawing predominantly on OA and other free-to-read full-text articles and reports. The findings reveal that genAI is making OA outputs central to criminology's evidence base. Free materials are easier for both humans and machines to discover, scrutinize, and integrate into policy and practice. Paywalled research is more likely to be ignored. The takeaway is clear: criminologists who keep their work closed access are blocking its inclusion in the evidence base.

Introduction

Open access (OA) to criminology outputs—articles, data, code, and other scholarly resources—is essential for building a solid evidence base. An output is OA if it is "digital, online, free of charge, and free of most copyright and licensing restrictions" (Suber, n.d.). Making criminology outputs OA maximizes their uptake and, in turn, supports better informed policy decisions, more effective crime-prevention strategies, reduced injustices in the legal system, and accelerated scientific progress.

The case for OA is even more compelling in light of generative artificial intelligence (genAI), especially large language models (LLMs)—systems that generate content from natural-language prompts (Stryker and Scapicchio, 2024; Yu and Guo, 2023). Although first developed just to guess the next word in a string of text, these models now perform a wide range of complex tasks (Kelleher, 2019; Raaijmakers, 2025; Wheeler, 2026).

GenAI has quickly become a key tool for research discovery, review, and synthesis. Information users—including researchers, policymakers, practitioners, students, and the general public—increasingly rely on this new form of intelligence to locate, evaluate, interpret, and apply scholarly knowledge.

The boom in genAI began in 2022. Until then, genAI tools were effectively limited to technically skilled users, such as those able to train models.[1] With the release of ChatGPT (OpenAI, n.d.a) and the rapid emergence of competitors such as Gemini (Google, n.d.) and Perplexity (n.d.), genAI entered a new era of widespread accessibility and everyday use.

By introducing chat-style interfaces, genAI no longer requires users to have technical expertise. With simple natural-language prompts, anyone with an internet-connected computer or smart device can use genAI to identify, digest, summarize, and brainstorm applications of scholarly literature.

Like humans, the ability of genAI to produce accurate, complete, and otherwise desirable outputs reflects their access to information. This is why OA outputs are essential for building a solid evidence base. By expanding what can be used, OA improves the discovery, review, synthesis, and application of scholarly knowledge. It does this by removing cost and legal barriers that prevent people and systems from finding, reading, and reusing the full content of scholarly outputs (Suber, n.d.; Willinsky, 2022).

Conversely, closed access constrains what both human minds and genAI can do. An output is closed access when it is unpublished, paywalled, or otherwise restricts who or what can use it and how (Jacques, 2023). These restrictions increase the likelihood that tools will offer lower-quality responses. If LLMs cannot see the material, it can neither be used in training future iterations of the model, or be referenced directly when the models search for additional materials.

Most criminology datasets, code libraries, and teaching materials remain locked away in private files. The bulk of our published articles require payment to access (Ashby, 2021). This is not out of necessity but choice (Jacques and Piza, 2022; Jacques and Wheeler, 2025; Piza and Jacques, 2020). Criminologists can make all of their outputs OA, whether by funding gold OA via an article processing charge (APC), choosing diamond OA venues with no APC, or providing green OA by posting preprints and postprints on personal webpages and in repositories such as CrimRxiv (n.d.) (Jacques, 2025a).

Criminologists are slowly but steadily joining the OA movement (e.g., CrimRxiv Consortium, n.d., 2025; European Network for Open Criminology, n.d.). This shift advances social justice by making our outputs available to everyone and strengthens science by rendering them more transparent, readily accessible, and reusable. The result is greater impact and higher return-on-investment (ROI)—more citations, pageviews, and practical applications for equal or lower investments of time, effort, and money (Jacques, 2025b; Worrall and Wilds, 2024).

GenAI is accelerating the salience of OA for criminology. These systems decide which sources to treat as part of the evidence base and—as we show below—disproportionately draw on OA and other free-to-read materials. This is unsurprising, since openly available, crawlable texts are easier for genAI to ingest, index, and reuse than closed-access content. The implication for authors is clear: if you want genAI to treat your work as part of the evidence base, you must ensure it has full access to your publications.

Deep-research sources

Users of popular genAI platforms are often presented with two options: a basic “search” alongside “deep research.” Deep research refers to a mode in which the model systematically scans the web, visits many sources, and assembles a structured, report-style answer to a prompt. In practice, this mode approximates what scholars would recognize as a literature review: it identifies relevant sources, extracts key findings, and organizes them into an integrated narrative.

We were curious how these tools would handle criminology literature reviews, so we ran a small informal study.[2] We used three platforms—Google's (n.d.) Gemini, OpenAI's (n.d.a) ChatGPT, and Perplexity (n.d.)[3]—and gave each a different criminology-related prompt: measuring stress in police officers (Gemini), the effectiveness of gunshot detection (ChatGPT), survey measures of attitudes toward police (Perplexity), and making convenience samples more representative (Perplexity).[4] The full deep-research reports for each query are available as online supplements (linked above), and we draw on them below to examine what these systems actually cite.

Google's Gemini and Perplexity produced usable reference lists by default, whereas OpenAI's ChatGPT required repeated prompting to generate them. In other words, Gemini and Perplexity preemptively disclose where their information comes from, while ChatGPT must be explicitly instructed to reveal its sources, placing an additional burden on users who want to check those sources.

What did these tools actually cite? Below, we examine the types of sources and the kinds of access to them. Specifically, we classify each citation as real or hallucinated; academic or non-academic; linked to freely available full-text or only to a summary; when full-text is freely available, whether it is "really OA" or merely "bronze access"—free to read at the time but with no legal guarantee it will remain so.

Real vs. hallucinated

The good and bad news about genAI tools is that they almost always produce answers that sound plausible, especially to uncritical readers. Yet their outputs vary widely in accuracy, completeness, and other features—such as citation practices, structure, and tone—depending on the model's training, the task and domain, the prompts used, and the size of the context window.

Like humans, genAI will sometimes make up an answer—"hallucinate"—rather than acknowledge they do not know (Li et al., 2025; Lowe, 2025; Wu et al., 2025). They can also misinterpret data, rely on outdated information, or draw on flawed sources, including findings from retracted studies. For both human-created and genAI-created content, users have a duty to check the claims and adjust their decisions accordingly. GenAI has not made critical thinking more important, but it has made the need for it more visible.

Unlike the reports from Google and Perplexity, OpenAI's reference list was essentially one large hallucination. It lists 12 works (shown in the above-linked PDF and in table 1), but every link fails. Link rot is a genuine problem for websites, yet it does not explain these results: these URLs never worked because the cited items were never real web pages (as the subsequent report shows). Instead, the references are portmanteaus[5] of real works and fabricated details.

To help make sense of these hallucinated references—and to illustrate how much platforms' reports can vary in quality—we asked Perplexity's deep-research model to analyze the citations in ChatGPT's bibliography. The full thread from this exercise is available as a linked PDF (linked above), which includes a summary table of Perplexity's assessment. For illustration, table 1 presents Perplexity's evaluation of the first five citations in ChatGPT's list (which we manually reviewed and it is accurate.)

The ChatGPT example underscores a central risk of relying on genAI for literature reviews: even when the narrative sounds plausible, the underlying references may be fabricated or corrupted, and users have no guarantee that the cited works actually exist. Users must therefore scrutinize the cited sources, which is also good practice when evaluating and building on human-generated literature reviews.

Table 1. Perplexity's analysis of ChatGPT's first five references

Citation (as given)	Real?	What's wrong?	Closest real source?
Mares, D., & Blackburn, E. (2021). *Evaluating the Impact of ShotSpotter Technology on Firearm Homicides and Arrests*. Criminal Justice Policy Review. https://doi.org/10.1177/0887403421992907	Partially hallucinated	Wrong title, wrong journal, wrong DOI, but correct authors/topic/year.	Real article: Mares, D., & Blackburn, E. (2021). *Acoustic gunshot detection systems: A quasi-experimental evaluation in St. Louis, MO. Journal of Experimental Criminology*, 17(2), 193–215. https://doi.org/10.1007/s11292-019-09405-x
National Institute of Justice. (2022). *Acoustic Gunshot Detection: A Review of the Evidence*. https://nij.ojp.gov/library/publications/acoustic-gunshot-detection-review-evidence	Likely hallucinated / misattributed	URL does not resolve; no NIJ report with this exact title; NIJ/OJP has related GDT reports around 2019–2022 but not this one.	Closest: BJA/NIJ/OJP materials such as *Gunshot Detection: Reducing Gunfire through Acoustic Technology* (BJA, 2022) and *Implementing Gunshot Detection Technology: Recommendations for Law Enforcement and Municipal Partners* (Urban Institute/NIJ, 2019–2020).
Piza, E. L., et al. (2023). *Gunshot Detection Technology and Police Response: A Quasi-Experimental Evaluation*. *Journal of Experimental Criminology*. https://doi.org/10.1007/s11292-023-09553-z	Partially hallucinated	Wrong DOI and inexact title; authors/year/journal/topic roughly right but conflated.	Real JEC article: Piza, E. L., Arietti, R. A., Carter, J. G., & Mohler, G. O. (2023). *The effect of gunshot detection technology on evidence collection and case clearance in Kansas City, Missouri. Journal of Experimental Criminology*. DOI ≈ 10.1007/s11292-023-09594-6 (not 09553-z).
Dallas Police Department. (2015). *ShotSpotter Evaluation Report*. https://dallaspolice.net/Reports/shotspotter2015.pdf	Hallucinated	URL and title do not exist; no 2015 DPD ShotSpotter evaluation report online.	Closest: Mazerolle, L. G., Watkins, C., Rogan, D., & Frank, J. (1998). *Using gunshot detection systems in police departments: The impact on police response times and officer workloads. Police Quarterly*, 1(2), 21–49 – based in Dallas but not a DPD report.

That Doesn't Stop Crime? https://www.govtech.com/public-safety/shotspotter-technology-that-doesnt-stop-crime	Partially hallucinated	Wrong year and title; URL given does not match any article.	Closest: GovTech, 2024. *Study: ShotSpotter Doesn't Reduce Crime or Shootings* (May 2, 2024) and *Chicago Latest City to Rethink Gunshot Detection Technology* (Feb 28, 2024).

Academic vs. non-academic

Conventional academic literature reviews typically restrict their citations to scholarly articles, books, reports, and conference proceedings. By contrast, deep research defaults to something closer to Wikipedia's citation practice (Reagle, 2010): a factual claim can be supported by almost any source—news articles, blogs, social media posts, and crowdsourced encyclopedias—so long as there is a clickable link to a specific item (e.g., OpenAI, n.d.b).

In our small study, however, the deep-research tools mostly drew on academic sources—or, in ChatGPT's case, on sources that appeared academic but were in fact fabricated. To illustrate the emphasis on academic sources, we now focus on Gemini's report on officer stress. Table 2 summarizes the first ten citations from that report, out of 78 total references.

All of the first ten sources are academic journal articles or reports.[6] Across the full set of 78 citations, the few non-academic sources consist mainly of internal procedure documents, policy guidelines, and materials on how to conduct evaluations (e.g., the Maslach Burnout Inventory, PTSD Checklist for DSM-5). These non-academic sources are appropriate to include in a literature review because they define key constructs, operationalize measures, and shape how stress and related outcomes are assessed in practice.

This example suggests that deep-research tools can surface the kinds of peer-reviewed studies that anchor the evidence base, while also drawing in practice documents that are relevant. Our caution is that, in addition to checking whether sources are real, users should watch for cases where these tools drift toward lower-quality or advocacy-driven material (for examples, see Johnson and Johnson, 2026).

Table 2. Source-type and access to the first ten cited works in Gemini's report on officer stress

Abbreviated title	Abbreviated URL	Academic source?	Full-text at the URL?	OA license or public domain?
Emergency Responder Exhaustion ...	https://www.frsn.org/uploads/1/3/1/2/131294856/ …	Yes	Yes	No
Work-related stress and psychological ...	https://pmc.ncbi.nlm.nih.gov/articles/PMC10970020	Yes	Yes	Yes
Best practices for counseling first ...	https://www.counseling.org/docs/default-source…	Yes	Yes	No
Job stress, burnout and coping in ...	https://www.mdpi.com/1660-4601/17/18/6718	Yes	Yes	Yes
Burnout and post-traumatic stress ...	https://www.researchgate.net/publication/3856790 …	Yes	Yes	Yes
Mental disorders and mental healt ...	https://pmc.ncbi.nlm.nih.gov/articles/PMC108751 …	Yes	Yes	Yes
SAMHSA disaster technical assistance ...	https://www.samhsa.gov/sites/default/files/dtac/s …	Yes	Yes	Yes
Helping the helpers	https://www.firstnet.com/content/dam/firstnet/wh …	Yes	Yes	No
A systematic review of the current ...	https://pmc.ncbi.nlm.nih.gov/articles/PMC68855 …	Yes	Yes	Yes
A cross-sectional analysis of traumatic ...	https://www.jems.com/mental-health-wellness/tra…	Yes	Yes	No

Free full-text vs. summary

Typically, academic literature reviews only draw on published sources.[7] What they tend not to do is restrict citation to works with freely available full-text—the complete article, not only an abstract or snippet. As a result, any nuance in paywalled sources remains out of sight, limiting readers' ability to interrogate the evidence and undermining independent scrutiny.

Because most criminology articles are closed access (Ashby, 2021), it is reasonable to assume that the references in our literature are overwhelmingly to restricted content. By extension, one might expect deep-research citations to point mainly to paywalled sources as well. If that were the case, readers who rely on genAI for literature reviews would still be pushed toward sources they cannot inspect directly, reproducing the same barriers to scrutiny that characterize traditional, paywalled publishing.

Yet in the officer-stress report, deep research primarily draws on sources with full-text available at the cited URL. All of the first ten citations meet this standard. Across the 78 citations, roughly one in five URLs do not directly lead to full-text content. In some cases, the full-text is reachable with an extra click, while in others the page only links onward to paywalled versions. A few links are broken, likely reflecting the lag between when the reports were generated (summer 2025) and analyzed for this article (winter 2025/26).

Overall, the report still leans heavily toward directly accessible full-text sources. The tool appears to favor sources that can be read in their entirety. This orientation is good for knowledge because it lets readers and reviewers inspect the full arguments, methods, and results for themselves, rather than relying on secondhand summaries. It also makes it easier to detect errors, replicate analyses, and integrate findings across studies, strengthening criminology's evidence base.

Really OA or bronze access

All OA sources are free to read, but not all free sources are OA. To be "really OA" (Jacques, 2025a), a work must be permanently and legally free of charge and free of most restrictions on reuse (Suber, n.d.). This is ultimately a matter of copyright and licensing (Suber, 2016; Willinsky, 2022). By default, the creator of a work owns the copyright and therefore controls whether it is made public and, if it is published, how it is distributed and under what conditions others may reuse it.

The oldest form of OA is the public domain (Boyle, 2010). If a work is in the public domain, no one owns the copyright and it can be used in any way: shared or sold, adapted or remixed. Any creator can place their work in the public domain, but in practice—at least for criminology—the law usually does the work. This happens in two main ways: the copyright term expires, or the work was created by the government. In the United States, for example, criminology sources produced by federal agencies are in the public domain, and older works enter the public domain on a rolling basis once their copyright has expired.

In contemporary scholarly publishing, most works become OA through a license (Suber, n.d., 2016). A license shifts a work from "all rights reserved" to "some rights reserved" (with public-domain works effectively having "no rights reserved"). For academic literature, Creative Commons (n.d.) licenses are the most common type. These licenses spell out how a work may be reshared and adapted, including conditions such as "non-commercial" use and "share alike" redistribution.

There are no takebacks with OA. Once a work has a Creative Commons license or enters the public domain, it cannot be converted to a more restrictive status (except by changes in the law, which are rare). The work is permanently and legally free to read, share, and—depending on the license—adapt. By contrast, when there is no such legal guarantee, content that is temporarily free to read can later be placed behind a paywall. This kind of ephemeral free-to-read status is referred to as "bronze access" (for details, see Jacques, 2023, 2025a).

In the report on officer stress, as shown in table 2, six of the first ten citations are to really OA sources: five have a Creative Commons license, and the SAMHSA source is in the public domain. The other four are bronze-access sources, with free access currently available but not protected by law. The breakdown is roughly similar across the report's 78 citations.

Given that most criminology articles are closed access (Ashby, 2021), this pattern suggests that deep research is disproportionately drawing on sources that are really OA. For more than a decade, OA has had an established "citation advantage" (Suber, n.d., 2016; Willinsky, 2022; in criminology, see Worrall and Wilds, 2024). Now, this visibility and impact advantage is being multiplied by genAI.

Discussion

In our exploratory study, we compared how the deep-research tools of three genAI models summarized the literature on a few criminology topics. ChatGPT produced a seemingly plausible report, but its accompanying reference list was almost entirely hallucinated. By contrast, Gemini and Perplexity generated structured reports that mostly cited academic journal articles and reports, supplemented by appropriate practice documents such as guidelines and measurement tools. Their citations largely pointed to freely readable full-text sources, many of which are really OA because they are in the public domain or carry a Creative Commons license.

Notwithstanding the obvious limitations of our little study, the findings point to a straightforward implication: genAI makes open-access (and other freely accessible) outputs central to criminology's evidence base. They are easier for humans and genAI to ingest, interrogate, and reuse than closed-access content. Their lessons are more likely to be discovered, scrutinized, and incorporated into future research, policy, and practice. By contrast, when criminology outputs remain paywalled, people and machines are more likely to overlook relevant evidence or rely on incomplete, outdated, or lower-quality information.

It is more important than ever for authors to make their work OA. Criminologists have not done a good job of this to date (Ashby, 2021). The field and its stakeholders are held back by closed access, but it does not need to be this way. Authors who are willing and able can pay to make their versions-of-record gold OA. Personally, however, we have never done this because the ROI is better with diamond OA and green OA. These routes cost authors nothing yet work just as well (Jacques, 2025a; Jacques and Piza, 2022; Piza and Jacques, 2020). Criminologists who keep their work closed access are blocking its inclusion in the evidence base.

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Footnotes

1. Training involves feeding large collections of data into a statistical software system and adjusting its internal parameters so the model learns patterns. For improvement, training also may involve human feedback that labels outputs as better or worse based on their goals. The outcome of training is a model that can generate original content, like a summary of scholarly literature. ↩
2. Our article is an expanded and revised version of Wheeler (2025). For formal examinations of deep research, see Li et al. (2025) and Lowe (2025). ↩
3. Anthropic’s (n.d.) Claude was not evaluated because it requires a paid subscription for deep research. The tests were run in August of 2025. OpenAI had automatic model routing at the time of testing, with access to

GPT-5 and earlier model versions. Gemini had access to its 2.5 versions of model families at the time. We intentionally stuck to each platform's default interface rather than activating specialized modes, so as to reflect the experience of a typical user at the time of testing. While it is true that future model versions may improve this, we contend that the popular chat applications—as of now, without extensive modification—still tend to often produce hallucinated citations. ↩

4. We gave each platform a different topic rather than running the same topic through all three because our aim was to show, as a case study, how these tools handle criminology literature in general. Future work could do the more systematic comparison by using the same topic across platforms and models. ↩

5. “Portmanteaus” is a fancy word for mashups in which an LLM blends pieces of different real sources into a single nonexistent one. ↩

6. Some readers will think it is important to distinguish articles from reports on the assumption the former are peer-reviewed prior to publication but the latter are not. We do not take that approach because some reports are peer-reviewed and the peer-review process for articles is sometimes low quality. ↩

7. Occasionally, a human-generated output may cite unpublished information, like “personal correspondence” or a conference presentation. ↩